Okan Umut Özen

Software Engineer

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SUMMARY

Software engineering graduate from Bahçeşehir University and former data science intern at NTT DATA Business Solutions. I build complete projects across web, machine learning, games and embedded systems, including two live, tested browser tools in TypeScript and Svelte. English C1.

EDUCATION

B.Sc. Software Engineering, Bahçeşehir University 2022 – 2026
Capstone: RC car telemetry and remote control system.

EXPERIENCE

Data Science Intern, NTT DATA Business Solutions Mar 2025 – May 2025

- Supported data analysis, modeling and reporting processes.
- Developed a fake news detection project using Microsoft Azure.

PROJECTS

Yaw-Match | *TypeScript, Svelte, Vite, Vitest, Zod* Live demo · GitHub

- Converts sensitivity between games with 360° distance matching, cm/360, eDPI and FOV-aware conversions, running entirely in the browser.
- Built a games database with per-game yaw values, sources and trust labels; added shareable links, copyable config lines and unit tests.

Aim-Tune | *TypeScript, Svelte, Vite, Vitest* Live demo · GitHub

- Flick and tracking tests that recommend a sensitivity range in cm/360 for any game.
- Splits tracking misses into ahead of and behind the target to reduce reaction-time bias.

RC Car Telemetry System | *Python, Raspberry Pi, Arduino, TCP/IP, SQLite* GitHub

- Capstone project. Designed a 3-tier system (Arduino, Raspberry Pi, PC) communicating over USB serial and TCP/IP for real-time remote control.
- Processed 6 IR sensors with debouncing and 4 wheel encoders on the Raspberry Pi to stream live telemetry, and built a CustomTkinter and Matplotlib dashboard for post-session analysis.

Autonomous Highway Driving with Deep Q-Network | *Python, Stable-Baselines3, Gymnasium* GitHub

- Trained a DQN agent to drive through dense multi-lane traffic while avoiding collisions.
- Designed a reward balancing speed, lane efficiency and safety; built the training and evaluation pipeline.

Google Maps Business Finder | *Python, FastAPI, Playwright, Docker* GitHub

- Collects business listings for a keyword and location and enriches them with public contact details from each business's website.
- Asynchronous Playwright pipeline with live progress streamed over WebSockets; frontend and backend run as separate Docker Compose services.

Smart Restaurant Order System | *Java 17, design patterns, UML* GitHub

- Applied GoF patterns (Strategy, Factory, Observer) and SOLID principles to build an extensible order-processing engine, documented with UML class, sequence and use case diagrams.

Other projects: German Traffic Sign Classifier (OpenCV, HOG, SVM/KNN), Tri-Jump and The Last Ember (Unity, C#), Project Zomboid mods (Lua).

SKILLS AND CERTIFICATES

Programming languages: Python, TypeScript, Java, C#, C++, SQL, Lua

Frameworks and libraries: Svelte, FastAPI, Playwright, Unity, Stable-Baselines3, OpenCV, scikit-learn, Matplotlib

Tools: Git, Docker, SQLite, Microsoft Azure, Vite, Vitest, Raspberry Pi, Arduino

Certificates: Google Project Management: Foundations of Project Management, Project Initiation, Capstone (2023 – 2024)

Languages: English (C1), German (A1), Turkish (native)